Yuanwang Zhang | Curriculum Vitae

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RESEARCH INTEREST

Brain Image Processing, Multimodal Fusion, Computer-aided Diagnosis, Image Synthesis
Medical Image Analysis, Computer Vision

EDUCATION

ShanghaiTech University **Shanghai, China**
M.E. in Computer Science *September 2021 - June 2024 (expected)*

- GPA: 3.72/4.00
- TOFEL: 104
- Supervised by Prof. Dinggang Shen

Chongqing University **Chongqing, China**
B.E. in Computer Science *September 2017 - June 2021*

- GPA: 3.52/4.00
- Centesimal grade average: 86.3

RESEARCH EXPERIENCE

Multimodal Fusion for Brain Disease Diagnosis *2021 - present*

- Mainly focus on the fusion of multimodal imaging data (MRI and PET), and non-imaging data (demographic information, medical history and cognitive scores), for brain disease diagnosis.
- Collect and process a large-scale public dataset (including six different cohorts) with over 7000 subjects with multimodal neuroimaging and clinical data.
- Design algorithms following the clinical diagnostic procedure.
- Deal with incomplete multimodal data.
- Develop a multimodal guidance framework to enhance the performance of single-modal input.

Medical Image Translation *2022 – present*

- Mainly focus on synthesizing one superior but expensive modality from another inferior but cheap modality.
- Use GAN to synthesize multi-tracer PET from MRI.
- Use diffusion model to synthesize high quality 7T MRI from 3T MRI (T1w image and SWI).

PUBLICATIONS

- **Vector Quantized Multi-modal Guidance for Alzheimer's Disease Diagnosis Based on Feature Imputation** [link]
  Yuanwang Zhang, Kaicong Sun, Yuxiao Liu, Zaixin Ou, Dinggang Shen
  MICCAI Workshop on Machine Learning in Medical Imaging (MLMI), 2023.
- **Transformer-based Multimodal Fusion for Early Diagnosis of Alzheimer's Disease Using Structural MRI and PET** [link]
  Yuanwang Zhang, Kaicong Sun, Yuxiao Liu, Dinggang Shen
  IEEE International Symposium on Biomedical Imaging (ISBI), 2023.
- **Development and Fast Transferring of General Connectivity-based Diagnosis Model to New Brain Disorders with Adaptive Graph Meta-learner** [link]
  Yuxiao Liu, Mianxin Liu, Yuanwang Zhang, Dinggang Shen
  MICCAI, 2023.

- **Learning Hierarchical-Order Functional Connectivity Networks for Mild Cognitive Impairment Diagnosis** [link]
  Yuxiao Liu, Mianxin Liu, Yuanwang Zhang, Dinggang Shen
  IEEE International Symposium on Biomedical Imaging (*ISBI*), 2023.

OTHER WORKS AND PROJECTS

- **Synthesizing 7T MRI from Routine 3T MRI Via Score-based Diffusion Model Using Paired and Unpaired Data**
  Yuanwang Zhang, Caohui Duan, Jiameng Liu, Kaicong Sun, Xin Lou, Dinggang Shen
  Submitted to *MICCAI* 2024.
- **A Modality-Flexible Framework for Alzheimer's Disease Diagnosis Following Clinical Routine**
  Yuanwang Zhang, Kaicong Sun, Yuxiao Liu, Dinggang Shen
  Submitted to *IEEE Transactions on Medical Imaging* (under review).
- **Achieving Multi-modal Brain Disease Diagnosis Performance Using Only Single-Modality Images Through Generative AI**
  Kaicong Sun, Yuanwang Zhang, Jiameng Liu, Ling Yu, Yan Zhou, Fang Xie, Qihao Guo, Han Zhang, Qian Wang, Dinggang Shen
  Submitted to *Communication Engineering* (under review).
- **Amyloid-β Deposition Prediction with Large Language Model Driven and Task Oriented Learning of Brain Functional Networks**
  Yuxiao Liu, Mianxin Liu, Yuanwang Zhang, Yihui Guan, Qi-Hao Guo, Fang Xie, Dinggang Shen
  Submitted to *IEEE Transactions on Medical Imaging* (under review).
- **Automatic labeling of 3D Coronary Artery Based on Point Cloud**
  Course Project.
- **Parallel Classification for Cervical Cancer Screening**
  Course Project.

SELECTED COURSES

- **Postgraduate:**
  Medical Image Processing and Analysis, Principles of Magnetic Resonance Imaging, Deep Learning, Medical Big-Data and Artificial Intelligence, Computer Graphics, Scientific Writing, Scientific Presentation
- **Undergraduate:**
  Advanced mathematics, Linear Algebra, Discrete Mathematics, Data Structure, Circuit and Analog Electronics Technology, Critical Reading and Writing, Mathematical Model, Design and Analysis of Algorithms, Operating Systems, Computer Networks, Foundation of Machine Learning

REFERENCES

- **Dr. Dinggang Shen**
  Prof. and Founding Dean, School of BME
  ShanghaiTech University, Shanghai, China
  Email: dgshen@shanghaitech.edu.cn
- **Dr. Kaicong Sun**
  Research Assistant Professor, School of BME
  ShanghaiTech University, Shanghai, China
  Email: sunkc@shanghaitech.edu.cn
- **IDEA Lab at ShanghaiTech University**
  Link: https://idea.bme.shanghaitech.edu.cn/home/index